

DeFi Super-App

AMM + Lending + ERC4626 Vault + DAO + The Graph

125

TESTS PASSED

91.85%

SRC LINE COVERAGE

2 days

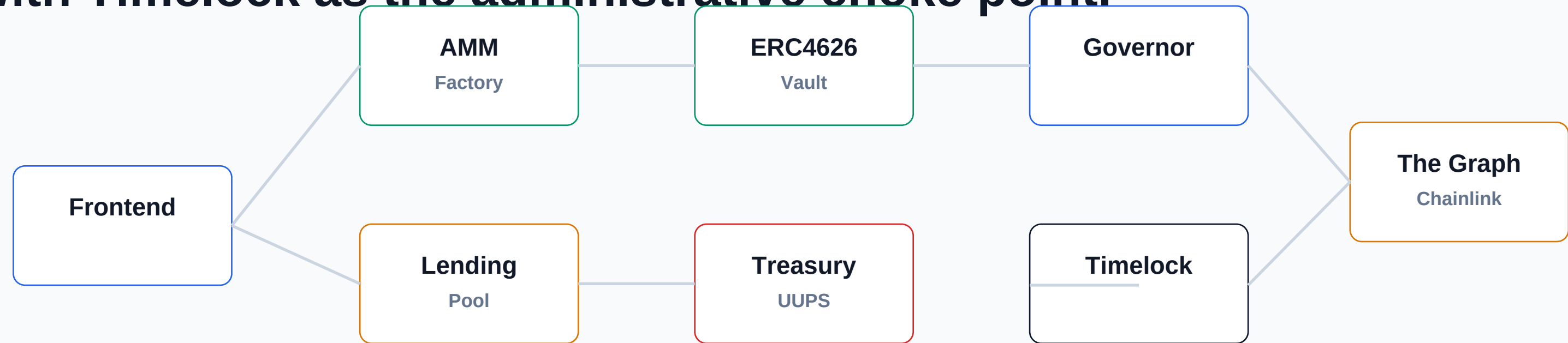
TIMELOCK DELAY

4%

GOVERNANCE QUORUM

Team: Tulegenov Alimzhan + Askhat Amirkhanov

The protocol is split into auditable primitives,
with Timelock as the administrative choke point.



A scratch-built constant-product pool covers swaps, slippage, LP shares, CREATE and CREATE2 deployment.

Core mechanics

$x * y = k$ invariant

0.3% swap fee

minAmountOut slippage guard

LP token is ERC20 share accounting

Factory proof

createPool uses CREATE

createPool2 uses CREATE2 assembly

predictAddress supports deterministic integrations

The lending pool adds the missing Option A primitive with price-aware risk checks.

50%

LTV cap

75%

liquidation threshold

5%

linear APR

5%

liquidation bonus

Flow: deposit collateral -> borrow USDC -> accrue interest -> repay or liquidation if health factor $< 1e18$.

Oracle isolation: lending uses Chainlink-backed PriceFeed adapters, not AMM spot prices.

ERC4626 yield accounting and UUPS Treasury upgrades are deliberately separated.

YieldVault

- ERC4626 shares
- SafeERC20 asset movement
- Owner-gated yield deposits
- Rounding behavior tested

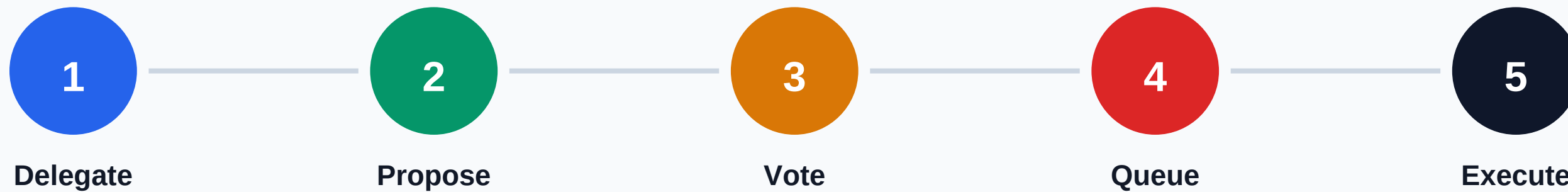
Treasury V1 -> V2

- UUPS proxy
- Timelock owner
- V2 appends paused flag
- Storage collision avoided

ERC721

- ProtocolBadge
- Timelock-owned
- Token standard checkbox
- Demo rewards

The full propose-vote-queue-execute path is tested against a Treasury withdrawal.



Parameters: 1 day voting delay, 1 week voting period, 4% quorum, 1% proposal threshold, 2 day Timelock.

The project is defended by layered testing, CEI, SafeERC20, Timelock control and CI checks.

Unit

93

public functions and revert paths

Fuzz

12

AMM, vault, voting power

Invariant

7

k, accounting, supply

Fork

4

USDC, Chainlink, Uniswap

The remaining live-demo step is a fresh Arbitrum Sepolia deployment with updated addresses.

1. Deploy and verify

Run Deploy.s.sol with PRIVATE_KEY, ARBITRUM_SEPOLIA_RPC, ARBISCAN_API_KEY and CHAINLINK_USDC_USD.

2. Verify and demo

Run VerifyDeployment.s.sol, set frontend env vars, deploy subgraph, then demonstrate swap, vault deposit and governance vote.